

An Infinite Pell-Type Family for

$$y^2 + x^2y + xz^2 + y - 1 = 0$$

Abstract

We give a completely explicit infinite family of integer solutions to

$$y^2 + x^2y + xz^2 + y - 1 = 0.$$

All solutions in the family have the fixed coordinate $x = -55$. The proof is an elementary Pell-type norm argument: one seed point is propagated by an integral linear transformation preserving the quadratic form $U^2 - 55V^2$.

1 The infinite family

Define integer sequences $(U_n)_{n \geq 0}$ and $(V_n)_{n \geq 0}$ by

$$U_0 = 1575, \quad V_0 = 59, \tag{1}$$

and

$$\begin{pmatrix} U_{n+1} \\ V_{n+1} \end{pmatrix} = \begin{pmatrix} 89 & 660 \\ 12 & 89 \end{pmatrix} \begin{pmatrix} U_n \\ V_n \end{pmatrix}; \tag{2}$$

that is,

$$U_{n+1} = 89U_n + 660V_n, \quad V_{n+1} = 12U_n + 89V_n.$$

Equivalently,

$$U_n + V_n\sqrt{55} = (1575 + 59\sqrt{55})(89 + 12\sqrt{55})^n. \tag{3}$$

Lemma 1 (Norm preservation). *For all integers U, V , if*

$$U' = 89U + 660V, \quad V' = 12U + 89V,$$

then

$$(U')^2 - 55(V')^2 = U^2 - 55V^2.$$

Proof. Direct expansion gives

$$\begin{aligned} (U')^2 - 55(V')^2 &= (89U + 660V)^2 - 55(12U + 89V)^2 \\ &= (89^2 - 55 \cdot 12^2)U^2 + 2 \cdot 89(660 - 55 \cdot 12)UV \\ &\quad + (660^2 - 55 \cdot 89^2)V^2. \end{aligned}$$

The three coefficients are respectively

$$89^2 - 55 \cdot 12^2 = 1, \quad 2 \cdot 89(660 - 55 \cdot 12) = 0, \quad 660^2 - 55 \cdot 89^2 = -55.$$

Hence the displayed expression is $U^2 - 55V^2$. □

Theorem 1. For every $n \in \mathbb{Z}_{\geq 0}$ and every $\varepsilon, \delta \in \{\pm 1\}$,

$$(x, y, z) = (-55, \varepsilon U_n - 1513, \delta V_n) \quad (4)$$

is an integer solution of

$$y^2 + x^2 y + x z^2 + y - 1 = 0. \quad (5)$$

Moreover, the triples with $\varepsilon = \delta = 1$ are pairwise distinct. Consequently, (5) has infinitely many integer solutions.

Proof. First,

$$1575^2 - 55 \cdot 59^2 = 2\,480\,625 - 191\,455 = 2\,289\,170 = 1513^2 + 1. \quad (6)$$

Lemma 1, applied inductively to (2), therefore yields

$$U_n^2 - 55V_n^2 = 1513^2 + 1 \quad (n \geq 0). \quad (7)$$

Now fix $n \geq 0$ and $\varepsilon, \delta \in \{\pm 1\}$, and set

$$x = -55, \quad y = \varepsilon U_n - 1513, \quad z = \delta V_n.$$

Because $x^2 + 1 = 3026 = 2 \cdot 1513$, the left-hand side of (5) is

$$\begin{aligned} y^2 + x^2 y + x z^2 + y - 1 &= y^2 + 3026y - 55z^2 - 1 \\ &= (y + 1513)^2 - 55z^2 - (1513^2 + 1) \\ &= U_n^2 - 55V_n^2 - (1513^2 + 1) = 0 \end{aligned}$$

by (7). Thus every triple in (4) is an integer solution.

It remains only to prove that infinitely many of these triples are distinct. Since $U_0, V_0 > 0$ and every entry of the recurrence matrix in (2) is positive, induction gives $U_n, V_n > 0$ for all n . Hence

$$V_{n+1} = 12U_n + 89V_n > V_n.$$

Thus $(V_n)_{n \geq 0}$ is strictly increasing, so the triples obtained with $\varepsilon = \delta = 1$ are pairwise distinct. $\square$

For example, the first three members of this subfamily are

$$(-55, 62, 59), \quad (-55, 177602, 24151), \quad (-55, 31879382, 4298819).$$

The independent sign changes in (4) reflect the symmetries

$$z \mapsto -z, \quad y \mapsto -3026 - y$$

of the equation after x has been fixed at -55 .

2 How the construction was found

The main obstacle in (5) is the mixed role of x : it occurs both as the coefficient of z^2 and, through x^2 , as a large coefficient of y . Choosing x negative turns the equation into an indefinite quadratic equation in (y, z) , which is precisely the setting in which Pell-type propagation becomes available.

Write

$$x = -m, \quad m > 0.$$

Then (5) becomes

$$y^2 + (m^2 + 1)y - mz^2 - 1 = 0. \quad (8)$$

For odd m , completing the square gives

$$\left(y + \frac{m^2 + 1}{2}\right)^2 - mz^2 = \left(\frac{m^2 + 1}{2}\right)^2 + 1. \quad (9)$$

Thus, once one finds a single integral point for a suitable nonsquare m , a solution of the Pell equation $r^2 - ms^2 = 1$ can be used to generate further points on the same norm level.

The question is therefore how to manufacture a convenient seed. The dominant terms in (8) suggest looking for y and z close to m . Put

$$y = m + a, \quad z = m + b.$$

Substitution into the left-hand side of (8) yields

$$(a + 1 - 2b)m^2 + (2a - b^2 + 1)m + (a^2 + a - 1). \quad (10)$$

The cubic terms cancel automatically. We can also force the quadratic term to vanish by taking

$$a = 2b - 1.$$

Then (10) reduces to

$$-(b^2 - 4b + 1)m + 4b^2 - 2b - 1.$$

Consequently, a seed is obtained whenever

$$m = \frac{4b^2 - 2b - 1}{b^2 - 4b + 1} \quad (11)$$

is a positive integer.

A particularly natural way to remove the divisibility issue in (11) is to make the denominator equal to 1. Since

$$b^2 - 4b + 1 = (b - 2)^2 - 3,$$

the equation $b^2 - 4b + 1 = 1$ gives $(b - 2)^2 = 4$, hence $b = 0$ or $b = 4$. The choice $b = 0$ produces $m = -1$, whereas $b = 4$ gives

$$m = 4 \cdot 4^2 - 2 \cdot 4 - 1 = 55, \quad a = 2 \cdot 4 - 1 = 7.$$

Therefore

$$x = -55, \quad y = 55 + 7 = 62, \quad z = 55 + 4 = 59,$$

which is the seed solution appearing in Theorem 1.

For $m = 55$, equation (9) becomes

$$(y + 1513)^2 - 55z^2 = 1513^2 + 1.$$

The small identity

$$89^2 - 55 \cdot 12^2 = 1$$

then supplies a norm-one multiplier. If $U = y + 1513$ and $V = z$, multiplication by $89 + 12\sqrt{55}$ sends

$$U + V\sqrt{55} \quad \text{to} \quad (89U + 660V) + (12U + 89V)\sqrt{55},$$

which is exactly the recurrence (2). In this way, the seed point and the norm-one identity together explain every coefficient in the final infinite family.

Remark. Theorem 1 proves infinitude, as required; it does not assert that the displayed family exhausts all integer solutions of (5).